Supervyse OPF Redfish Network Card Information

Revision 1.0
January 2025

Copyright (c) 2024, All Rights Reserved. Insyde Software.

No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form, or by any means, electronic, mechanical, magnetic, optical, chemical, manual or otherwise, without the prior written permission of Insyde Software Corp.

Disclaimer

Insyde Software provides this document and the programs "as is" without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranties of merchantability or fitness for a particular purpose.

This document could contain technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in future revisions of this document. Insyde Software is under no obligation to notify any person of the changes.

The following trademarks are used in this document:

Insyde and InsydeH2O are registered trademarks or trademarks of Insyde Software. All other trademarks or trade names are property of their respective holders.

Contents

1 Introduction 6

1.1 Network card device 6

1.2 Rdfish Schema 6

2 Schema Example 9

2.1 NetworkInterfaceCollection 9
2.2 NetworkInterface 9
2.3 NetworkAdapterCollection 10
2.4 NetworkAdapter 10
2.5 NetworkAdapterMetrics 11
2.6 NetworkDeviceFunctionCollection 11
2.7 NetworkDeviceFunction 12
2.8 NetworkDeviceFunctionMetrics 13
2.9 PortCollection 13
2.10 Port 14
2.11 PortMetrics 14
2.12 InsydeNcsiCollection 15
2.13 InsydeNcsi 16
2.14 InsydeNcsiPackage 16
2.15 Common metrics value comparison table 23
2.16 InsydePCIeFunctions 25

Revision History

Revision Number	Description	Author	Reviewer	Release Date
1.0	Initial release.	Jolin Tsai	Victor Wu	January 2025

1 Introduction

This document explains how to retrieve network card information using Redfish.

1.1 Network card device

We support three type of network card interfaces.

- MCTP over RMII
- MCTP over PCIe
- MCTP over SMBus

To prevent capturing duplicate devices that support multiple interfaces, we use the MAC address as a unique identifier to determine if the device is the same.

If a device is detected on multiple interfaces, the following priority is applied:

1. MCTP over RMII
2. MCTP over PCIe and MCTP over SMBus

The device type is display in InsydeNcsi schema property "DeviceType".

- NCSIOverRBT
- PCIe
- SMBus

1.2 Rdfish Schema

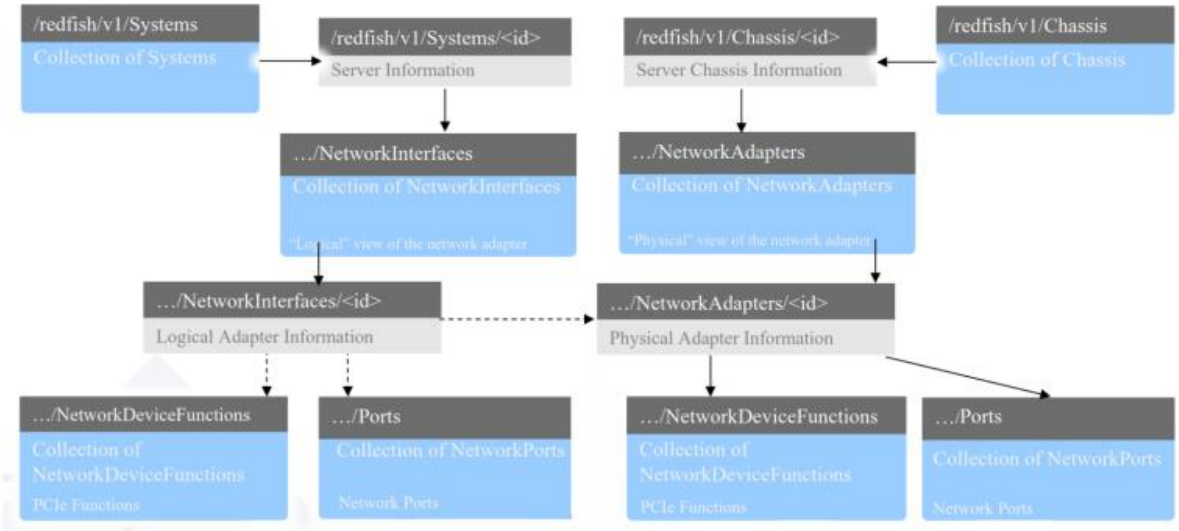
We have several schema that capture network card information in Redfish:

Standard DMTF schema:

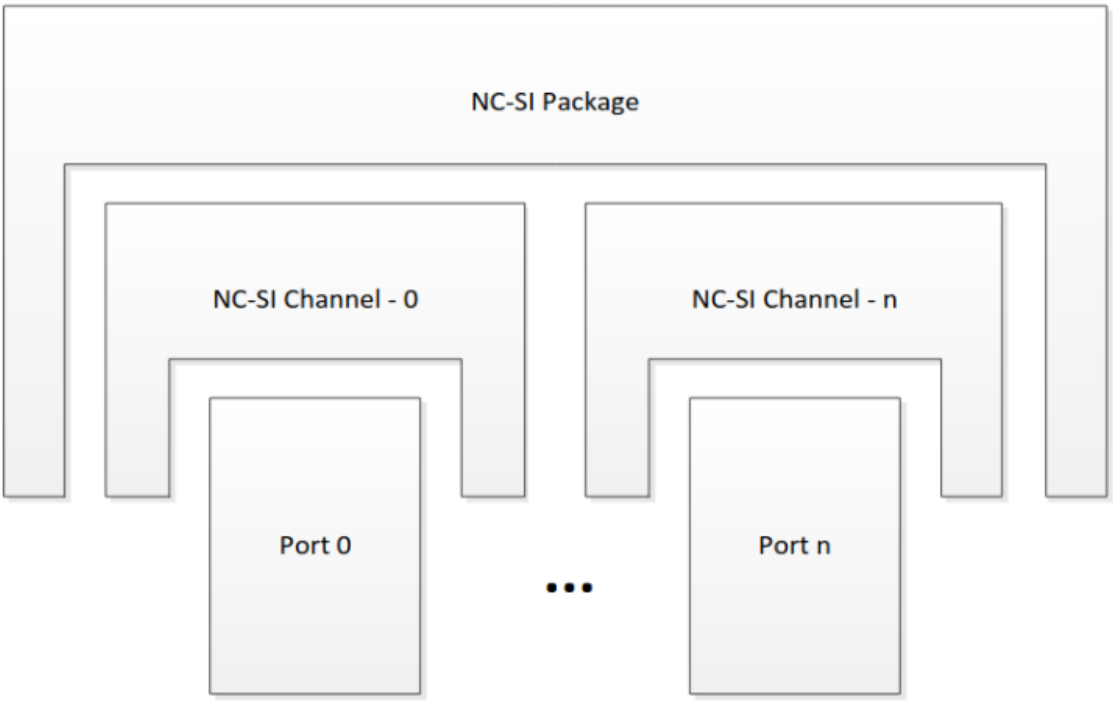
- NetworkInterfaceCollection
- NetworkInterface
- NetworkAdapterCollection
- NetworkAdapter
- NetworkAdapterMetrics
- NetworkDeviceFunctionCollection
- NetworkDeviceFunctions
- NetworkDeviceFunctionMetrics
- Port
- PortMetics

Insyde OEM schema:

- InsydeNcsi: include the device type and controller-related version information
- InsydeNcsiPackage: include the channel mac, current link status information and packet statistics



A channel is associated with a network port and a package is usually associated with a physical device that exposes a single NC-SI device.



- Use NCSI device index as token: NetworkInterface, NetworkAdapter, InsydeNcsi
- Use NCSI Package id as token: InsydeNcsiPackage
- Use NCSI Channel index as token: NetworkDeviceFunction, Port

2 Schema Example

2.1 NetworkInterfaceCollection

URI: /redfish/v1/Systems/system/NetworkInterfaeces


```
{
   "@odata.context":
"/redfish/v1/$metadata#NetworkInterfaceCollection.NetworkInterfaceCollection",
   "@odata.id": "/redfish/v1/Systems/system/NetworkInterfaces",
   "@odata.type": "#NetworkInterfaceCollection.NetworkInterfaceCollection",
   "Description": "The Collection of Network Interfaces",
   "Members": [
     {
       "@odata.id": "/redfish/v1/Systems/system/NetworkInterfaces/1"
     },
     {
       "@odata.id": "/redfish/v1/Systems/system/NetworkInterfaces/2"
     }
   ],
   "Members@odata.count": 2,
   "Name": "NetworkInterface Collection"
 }
```


2.2 NetworkInterface

URI: /redfish/v1/Systems/system/NetworkInterfaces/{ncsi_index}


```
{
   "@odata.context": "/redfish/v1/$metadata#NetworkInterface.NetworkInterface",
   "@odata.id": "/redfish/v1/Systems/system/NetworkInterfaces/1",
   "@odata.type": "#NetworkInterface.v1_2_0.NetworkInterface",
   "Description": "NetworkInterface",
   "Id": "1",
   "Links": {
     "NetworkAdapter": {
       "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1"
     }
   },
   "Name": "1",
   "NetworkDeviceFunctions": {
     "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions"
   },
   "Ports": {
     "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/Ports"
   },
   "Status": {
     "Health": "OK",
     "HealthRollup": "OK",
     "State": "Enabled"
```

```
    }
  }
```

2.3 NetworkAdapterCollection

URI: /redfish/v1/Chassis/{chassis_id}/NetworkAdapters


```
{
    "@odata.context":
"/redfish/v1/$metadata#NetworkAdapterCollection.NetworkAdapterCollection",
    "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters",
    "@odata.type": "#NetworkAdapterCollection.NetworkAdapterCollection",
    "Description": "The NetworkAdapterCollection schema describes a collection of network
adapter instances.",
    "Members": [
        {
            "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1"
        },
        {
            "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2"
        }
    ],
    "Members@odata.count": 2,
    "Name": "Network Adapter Collection"
}
```


2.4 NetworkAdapter

URI: /redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}


```
{
   "@odata.context": "/redfish/v1/$metadata#NetworkAdapter.NetworkAdapter",
   "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2",
   "@odata.type": "#NetworkAdapter.v1_8_0.NetworkAdapter",
   "Description": "A NetworkAdapter represents the physical network adapter capable of
connecting to a computer network.",
   "Id": "2",
   "Manufacturer": "0x0157",
   "Metrics": {
     "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/Metrics"
   },
   "Model": "0x1533",
   "Name": "2",
   "NetworkDeviceFunctions": {
     "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/NetworkDeviceFunctions"
   },
   "Ports": {
     "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/Ports"
   },
   "Status": {
     "Health": "OK",
```

```
      "HealthRollup": "OK",
      "State": "Enabled"
    }
  }
```

2.5 NetworkAdapterMetrics

URI: /redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/Metrics


```
{
 "@odata.context": "/redfish/v1/$metadata#NetworkAdapterMetrics.NetworkAdapterMetrics",
 "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/Metrics",
 "@odata.type": "#NetworkAdapterMetrics.v1_0_0.NetworkAdapterMetrics",
 "Description": "The NetworkAdapterMetrics schema contains usage and health statistics for a
network adapter.",
  "Actions": {
     "#NetworkDeviceFunctionMetrics.ResetMetrics": {
         "target":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions/1/Metrics/Action
s/NetworkDeviceFunctionMetrics.ResetMetrics"
        }
  },
 "Id": "NetworkAdapterMetrics",
 "NCSIRXFrames": 14,
 "NCSITXFrames": 13,
 "Name": "Network Adapter Metrics",
 "RXBytes": 0,
 "RXMulticastFrames": 0,
 "RXUnicastFrames": 0,
 "TXBytes": 0,
 "TXMulticastFrames": 0,
 "TXUnicastFrames": 0
}
```


- Action that Reset Metrics:

```
curl -k -u root:OpenBmc -X POST
https://{IP}/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/{ncsi_index}/Metrics/Actions/Ne
tworkDeviceFunctionMetrics.ResetMetrics
```

2.6 NetworkDeviceFunctionCollection

URI:
**/redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/NetworkDeviceFunctio
ns**

```
{
    "@odata.context":
"/redfish/v1/$metadata#NetworkDeviceFunctionCollection.NetworkDeviceFunctionCollection",
    "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions",
    "@odata.type": "#NetworkDeviceFunctionCollection.NetworkDeviceFunctionCollection",
    "Description": "The NetworkDeviceFunctionCollection schema describes a collection of
network device function instances.",
    "Members": [
        {
            "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions/1"
        },
        {
            "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions/2"
        },
        {
            "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions/3"
        },
        {
            "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/NetworkDeviceFunctions/4"
        }
    ],
    "Members@odata.count": 4,
    "Name": "Network Device Function Collection"
}
```


2.7 NetworkDeviceFunction

URI:
/redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/NetworkDeviceFunctions/{channel_index}


```
{
   "@odata.context": "/redfish/v1/$metadata#NetworkDeviceFunction.NetworkDeviceFunction",
   "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/NetworkDeviceFunctions/1",
   "@odata.type": "#NetworkDeviceFunction.v1_7_0.NetworkDeviceFunction",
   "Description": "The NetworkDeviceFunction schema represents a logical interface that a
network adapter exposes.",
   "Ethernet": {
     "MACAddress": "a0:36:9f:70:0f:7f",
     "PermanentMACAddress": "a0:36:9f:70:0f:7f"
   },
   "Id": "1",
   "Metrics": {
     "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/NetworkDeviceFunctions/1/Metrics"
   },
   "Name": "1",
   "Status": {
     "Health": "OK",
     "HealthRollup": "OK",
```

```
      "State": "Enabled"
    }
  }
```

2.8 NetworkDeviceFunctionMetrics

URI: /redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/NetworkDeviceFunctions/{channel_index}/Metrics

```
{
   "@odata.context":
"/redfish/v1/$metadata#NetworkDeviceFunctionMetrics.NetworkDeviceFunctionMetrics",
   "@odata.id":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/NetworkDeviceFunctions/1/Metrics",
   "@odata.type": "#NetworkDeviceFunctionMetrics.v1_1_0.NetworkDeviceFunctionMetrics",
   "Actions": {
     "#NetworkDeviceFunctionMetrics.ResetMetrics": {
       "target":
"/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/NetworkDeviceFunctions/1/Metrics/Action
s/NetworkDeviceFunctionMetrics.ResetMetrics"
     }
   },
   "Description": "The NetworkDeviceFunctionMetrics schema contains usage and health
statistics for a network function of a network adapter.",
   "Id": "NetworkDeviceFunctionMetrics",
   "Name": "Network Device Function Metrics",
   "RXBytes": 0,
   "RXFrames": 0,
   "RXMulticastFrames": 0,
   "RXUnicastFrames": 0,
   "TXBytes": 0,
   "TXFrames": 0,
   "TXMulticastFrames": 0,
   "TXUnicastFrames": 0
 }
```

- Action that Reset Metrics:

```
curl -k -u root:0penBmc -X POST
https://{IP}/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/{ncsi_index}/NetworkDeviceFun
ctions/{channel_index}/Metrics/Actions/NetworkDeviceFunctionMetrics.ResetMetrics
```

2.9 PortCollection

URI: /redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/Ports

```
{
   "@odata.context": "/redfish/v1/$metadata#PortCollection.PortCollection",
   "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/Ports",
   "@odata.type": "#PortCollection.PortCollection",
```

```
    "Members": [
        {
            "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/Ports/1"
        },
        {
            "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/Ports/2"
        },
        {
            "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/Ports/3"
        },
        {
            "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/1/Ports/4"
        }
    ],
    "Members@odata.count": 4,
    "Name": "Port Collection"
}
```


2.10 Port

URI:
/redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/Ports/{channel_index}


```
{
   "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/Ports/1",
   "@odata.type": "#Port.v1_9_0.Port",
   "Description": "The Port schema contains properties that describe a port of a switch,
controller,chassis, or any other device that could be connected to another entity.",
   "Ethernet": {
     "AssociatedMACAddresses": [
       "a0:36:9f:70:0f:7f"
     ],
     "FlowControlStatus": "None"
   },
   "Id": "1",
   "LinkConfiguration": [
     {
       "AutoSpeedNegotiationEnabled": true
     }
   ],
   "LinkStatus": "LinkDown",
   "Metrics": {
     "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/Ports/1/Metrics"
   },
   "Name": "1"
 }
```


2.11 PortMetrics

URI:
/redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/Ports/{channel_index}

```
{
   "@odata.id": "/redfish/v1/Chassis/BNC_Baseboard/NetworkAdapters/2/Ports/1/Metrics",
   "@odata.type": "#PortMetrics.v1_6_1.PortMetrics",
   "Actions": {
      "#PortMetrics.ResetMetrics": {
        "target":
"/redfish/v1/Chassis/PSU0/NetworkAdapters/2/Ports/1/Actions/PortMetrics.ResetMetrics"
      }
    },
   "Description": "The PortMetrics schema contains usage and health statistics for a switch
device or component port summary.",
   "Id": "Metrics",
   "Name": "Network Port Metrics",
   "Networking": {
     "RXFCSErrors": null,
     "RXFrames": 0,
     "RXMulticastFrames": 0,
     "RXPFCFrames": null,
     "RXPauseXOFFFrames": null,
     "RXPauseXONFrames": null,
     "RXUnicastFrames": 0,
     "TXFrames": 0,
     "TXMulticastFrames": 0,
     "TXMultipleCollisions": null,
     "TXPauseXOFFFrames": null,
     "TXPauseXONFrames": null,
     "TXSingleCollisions": null,
     "TXUnicastFrames": 0
   },
   "RXBytes": null,
   "TXBytes": null
 }
```


- Action that Reset Metrics:

```
curl -k -u root:0penBmc -X POST
https://{IP}/redfish/v1/Chassis/{chassis_id}/NetworkAdapters/{ncsi_index}/NetworkDeviceFunctio
ns/{channel_index}/Metrics/Actions/PortMetrics.ResetMetrics
```

2.12 InsydeNcsiCollection

URI: /redfish/v1/Systems/system/Oem/Insyde/Ncsi


```
{
    "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi",
    "@odata.type": "#InsydeNcsiCollection.InsydeNcsiCollection",
    "Description": "The NetworkAdapterCollection schema describes a collection of network
adapter instances.",
    "Members": [
        {
            "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi/1"
        },
        {
```

```
            "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi/2"
        }
    ],
    "Members@odata.count": 2,
    "Name": "Insyde Ncsi Collection"
}
```


2.13 InsydeNcsi

URI: /redfish/v1/Systems/system/Oem/Insyde/Ncsi/{ncsi_index}


```
{
    "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi/1",
    "@odata.type": "#InsydeNcsi.v1_0_0.InsydeNcsi",
    "Description": "The InsydeNcsi schema contains properties related to NCSI device.",
    "DeviceType": "PCIe",
    "Id": "1",
    "Name": "1",
    "Package": [
        {
            "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi/1/Package/0"
        }
    ],
    "VersionID": {
        "FirmwareName": "I210  FW ver",
        "FirmwareVersion": "01.07.00.03",
        "ManufacturerID": "0x0157",
        "NcsiVersion": "1.0.0",
        "PCIDID": "0x1533",
        "PCISSID": "0x0001",
        "PCISVID": "0x8086",
        "PCIVID": "0x8086"
    }
}
```


2.14 InsydeNcsiPackage

URI:
/redfish/v1/Systems/system/Oem/Insyde/Ncsi/{ncsi_index}/Package/{package_id}

Noted: Some properties may be null depending on the capability of the device

MCTP over RMII devices:


```
{
   "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi/1/Package/0",
   "@odata.type": "#InsydeNcsiPackage.v1_0_0.InsydeNcsiPackage",
   "Description": "The InsydeNcsiPackage schema contains properties related to
NcsiPackage.",
   "Id": "0",
```

```
    "Name": "0",
    "PackageInfo": [
        {
            "Capabilities": {
                "ARPPackets": "Forward To BMC",
                "Allmulticastaddressessupport": true,
                "AnyVLANnonVLAN": true,
                "BufferingCapability": 8192,
                "ConfigurationRequiredAENcontrol": true,
                "DHCPClientPackets": "Forward To BMC",
                "DHCPServerPackets": "Forward To BMC",
                "DHCPv6relayandservermulticast": "Forward To BMC",
                "HardwareArbitrationCapability": false,
                "HostNCDriverStatus": true,
                "HostNCDriverStatusChangeAENcontrol": true,
                "IPv6NeighborAdvertisement": "Forward To BMC",
                "IPv6RouterAdvertisement": "Forward To BMC",
                "LinkStatusChangeAENcontrol": true,
                "ManagementControllertoNetworkControllerFlowControlSupport": true,
                "MixedFilterCount": 2,
                "MulticastFilterCount": 0,
                "NetBIOSPackets": "Forward To BMC",
                "NetworkControllertoManagementControllerFlowControlSupport": true,
                "OEMSpecificAENControl": "0x2",
                "UnicastFilterCount": 0,
                "VLANFilterCount": 8,
                "VLANnonVLAN": false,
                "VLANonly": true
            },
            "ChannelIndex": 1,
            "LinkStatus": {
                "AutoNegotiateComplete": false,
                "AutoNegotiateFlag": true,
                "HostNCDriverStatusIndication": false,
                "Link": "Down",
                "LinkPartnerAdvertisedFlowControl": null,
                "LinkPartnerAdvertisedSpeed100T4": null,
                "LinkPartnerAdvertisedSpeedandDuplex1000TFD": null,
                "LinkPartnerAdvertisedSpeedandDuplex1000THD": null,
                "LinkPartnerAdvertisedSpeedandDuplex100TXFD": null,
                "LinkPartnerAdvertisedSpeedandDuplex100TXHD": null,
                "LinkPartnerAdvertisedSpeedandDuplex10TFD": null,
                "LinkPartnerAdvertisedSpeedandDuplex10THD": null,
                "OEMLinkSpeedValid": false,
                "ParallelDetectionFlag": false,
                "RXFlowControlFlag": false,
                "SerDesLink": false,
                "SpeedAndDuplex": "Auto-negotiate Not Complete",
                "TXFlowControlFlag": false
            },
            "MACAddress": "68:05:CA:9C:25:B9",
            "NcsiStatistics": {
                "AENsSent": 0,
                "NCSICommandChecksumErrors": 0,
                "NCSICommandTypeErrors": 0,
                "NCSICommandsReceived": 426,
                "NCSIControlPacketsDropped": 0,
                "NCSIReceivePackets": null,
                "NCSITransmitPackets": null
            },
```

```
            "NcsicontrollerPacketStatistics": {
                "1024To1522ByteFramesReceived": null,
                "1024To1522ByteFramesTransmitted": null,
                "128To255ByteFramesReceived": null,
                "128To255ByteFramesTransmitted": null,
                "1523To9022ByteFramesReceived": null,
                "1523To9022ByteFramesTransmitted": null,
                "256To511ByteFramesReceived": null,
                "256To511ByteFramesTransmitted": null,
                "512To1023ByteFramesReceived": null,
                "512To1023ByteFramesTransmitted": null,
                "64ByteFramesReceived": null,
                "64ByteFramesTransmitted": null,
                "65To127ByteFramesReceived": null,
                "65To127ByteFramesTransmitted": null,
                "AlignmentErrors": 0,
                "ControlFramesReceived": null,
                "CountersClearedFromLastRead": 123900,
                "ErrorJabberPacketsReceived": null,
                "ErrorRuntPacketsReceived": null,
                "ExcessiveCollisionFrames": null,
                "FCSReceiveErrors": 0,
                "FalseCarrierDetections": null,
                "JabberPacketsReceived": null,
                "LateCollisionFrames": null,
                "MultipleCollisionTransmitFrames": null,
                "PauseXOFFFramesReceived": 0,
                "PauseXOFFFramesTransmitted": 0,
                "PauseXONFramesReceived": 0,
                "PauseXONFramesTransmitted": 0,
                "RuntPacketsReceived": null,
                "SingleCollisionTransmitFrames": null,
                "TotalBroadcastPacketsReceived": 0,
                "TotalBroadcastPacketsTransmitted": 0,
                "TotalBytesReceived": null,
                "TotalBytesTransmitted": null,
                "TotalMulticastPacketsReceived": 0,
                "TotalMulticastPacketsTransmitted": 0,
                "TotalUnicastPacketsReceived": 0,
                "TotalUnicastPacketsTransmitted": 0,
                "ValidBytesReceived": null
            },
            "PassThroughStatistics": {
                "RXPacketChannelStateErrors": null,
                "RXPacketOversizedErrors": null,
                "RXPacketUndersizedErrors": null,
                "RXPacketsReceivedonLANInterface": 0,
                "TXPacketChannelStateErrors": null,
                "TXPacketOversizedErrors": null,
                "TXPacketUndersizedErrors": null,
                "TXPacketsDropped": null,
                "TXPacketsReceivedonNCSIInterface": 0,
                "TotalRXPacketsDropped": 0
            }
        },
        {
            "Capabilities": {
                "ARPPackets": "Forward To BMC",
                "Allmulticastaddressessupport": true,
                "AnyVLANnonVLAN": true,
```

```
            "BufferingCapability": 8192,
            "ConfigurationRequiredAENcontrol": true,
            "DHCPClientPackets": "Forward To BMC",
            "DHCPServerPackets": "Forward To BMC",
            "DHCPv6relayandservermulticast": "Forward To BMC",
            "HardwareArbitrationCapability": false,
            "HostNCDriverStatus": true,
            "HostNCDriverStatusChangeAENcontrol": true,
            "IPv6NeighborAdvertisement": "Forward To BMC",
            "IPv6RouterAdvertisement": "Forward To BMC",
            "LinkStatusChangeAENcontrol": true,
            "ManagementControllertoNetworkControllerFlowControlSupport": true,
            "MixedFilterCount": 2,
            "MulticastFilterCount": 0,
            "NetBIOSPackets": "Forward To BMC",
            "NetworkControllertoManagementControllerFlowControlSupport": true,
            "OEMSpecificAENControl": "0x2",
            "UnicastFilterCount": 0,
            "VLANFilterCount": 8,
            "VLANnonVLAN": false,
            "VLANonly": true
        },
        "ChannelIndex": 2,
        "LinkStatus": {
            "AutoNegotiateComplete": false,
            "AutoNegotiateFlag": true,
            "HostNCDriverStatusIndication": false,
            "Link": "Down",
            "LinkPartnerAdvertisedFlowControl": null,
            "LinkPartnerAdvertisedSpeed100T4": null,
            "LinkPartnerAdvertisedSpeedandDuplex1000TFD": null,
            "LinkPartnerAdvertisedSpeedandDuplex1000THD": null,
            "LinkPartnerAdvertisedSpeedandDuplex100TXFD": null,
            "LinkPartnerAdvertisedSpeedandDuplex100TXHD": null,
            "LinkPartnerAdvertisedSpeedandDuplex10TFD": null,
            "LinkPartnerAdvertisedSpeedandDuplex10THD": null,
            "OEMLinkSpeedValid": false,
            "ParallelDetectionFlag": false,
            "RXFlowControlFlag": false,
            "SerDesLink": false,
            "SpeedAndDuplex": "Auto-negotiate Not Complete",
            "TXFlowControlFlag": false
        },
        "MACAddress": "68:05:CA:9C:25:BA",
        "NcsiStatistics": {
            "AENsSent": 0,
            "NCSICommandChecksumErrors": 0,
            "NCSICommandTypeErrors": 0,
            "NCSICommandsReceived": 13,
            "NCSIControlPacketsDropped": 0,
            "NCSIReceivePackets": null,
            "NCSITransmitPackets": null
        },
        "NcsicontrollerPacketStatistics": {
            "1024To1522ByteFramesReceived": null,
            "1024To1522ByteFramesTransmitted": null,
            "128To255ByteFramesReceived": null,
            "128To255ByteFramesTransmitted": null,
            "1523To9022ByteFramesReceived": null,
            "1523To9022ByteFramesTransmitted": null,
```

```
                "256To511ByteFramesReceived": null,
                "256To511ByteFramesTransmitted": null,
                "512To1023ByteFramesReceived": null,
                "512To1023ByteFramesTransmitted": null,
                "64ByteFramesReceived": null,
                "64ByteFramesTransmitted": null,
                "65To127ByteFramesReceived": null,
                "65To127ByteFramesTransmitted": null,
                "AlignmentErrors": 0,
                "ControlFramesReceived": null,
                "CountersClearedFromLastRead": 123900,
                "ErrorJabberPacketsReceived": null,
                "ErrorRuntPacketsReceived": null,
                "ExcessiveCollisionFrames": null,
                "FCSReceiveErrors": 0,
                "FalseCarrierDetections": null,
                "JabberPacketsReceived": null,
                "LateCollisionFrames": null,
                "MultipleCollisionTransmitFrames": null,
                "PauseXOFFFramesReceived": 0,
                "PauseXOFFFramesTransmitted": 0,
                "PauseXONFramesReceived": 0,
                "PauseXONFramesTransmitted": 0,
                "RuntPacketsReceived": null,
                "SingleCollisionTransmitFrames": null,
                "TotalBroadcastPacketsReceived": 0,
                "TotalBroadcastPacketsTransmitted": 0,
                "TotalBytesReceived": null,
                "TotalBytesTransmitted": null,
                "TotalMulticastPacketsReceived": 0,
                "TotalMulticastPacketsTransmitted": 0,
                "TotalUnicastPacketsReceived": 0,
                "TotalUnicastPacketsTransmitted": 0,
                "ValidBytesReceived": null
            },
            "PassThroughStatistics": {
                "RXPacketChannelStateErrors": null,
                "RXPacketOversizedErrors": null,
                "RXPacketUndersizedErrors": null,
                "RXPacketsReceivedonLANInterface": 0,
                "TXPacketChannelStateErrors": null,
                "TXPacketOversizedErrors": null,
                "TXPacketUndersizedErrors": null,
                "TXPacketsDropped": null,
                "TXPacketsReceivedonNCSIInterface": 0,
                "TotalRXPacketsDropped": 0
            }
        }
    ]
}
```


MCTP over PCIe and MCTP over SMBus devices:


```
{
        "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/Ncsi/2/Package/0",
        "@odata.type": "#InsydeNcsiPackage.v1_0_0.InsydeNcsiPackage",
        "Description": "The InsydeNcsiPackage schema contains properties related to
NcsiPackage.",
        "Id": "0",
```

```
        "Name": "0",
        "PackageInfo": [
            {
                "Capabilities": {
                    "ARPPackets": "Forward To BMC",
                    "Allmulticastaddressessupport": true,
                    "AnyVLANnonVLAN": true,
                    "BufferingCapability": 8176,
                    "ConfigurationRequiredAENcontrol": true,
                    "DHCPClientPackets": "Forward To BMC",
                    "DHCPServerPackets": "Forward To BMC",
                    "DHCPv6multicastsfromservertoclientslisteningonwellknownUDPports":
"Filter Out",
                    "DHCPv6relayandservermulticast": "Forward To BMC",
                    "HardwareArbitrationCapability": false,
                    "HardwareArbitrationImplementationStatus": "Unknown",
                    "HostNCDriverStatus": true,
                    "IPv6MLD": "Filter Out",
                    "IPv6NeighborAdvertisement": "Forward To BMC",
                    "IPv6NeighborSolicitation": "Filter Out",
                    "IPv6RouterAdvertisement": "Forward To BMC",
                    "LinkStatusChangeAENcontrol": true,
                    "ManagementControllertoNetworkControllerFlowControlSupport": false,
                    "MixedFilterCount": 2,
                    "MulticastFilterCount": 0,
                    "NetBIOSPackets": "Forward To BMC",
                    "NetworkControllertoManagementControllerFlowControlSupport": false,
                    "UnicastFilterCount": 0,
                    "VLANFilterCount": 8,
                    "VLANnonVLAN": false,
                    "VLANonly": true
                },
                "ChannelIndex": 1,
                "LinkStatus": {
                    "AutoNegotiateComplete": true,
                    "AutoNegotiateFlag": true,
                    "HostNCDriverStatusIndication": true,
                    "Link": "Down",
                    "LinkPartnerAdvertisedFlowControl": "Both Symmetric & Asymmetric Pause",
                    "LinkPartnerAdvertisedSpeed100T4": false,
                    "LinkPartnerAdvertisedSpeedandDuplex1000TFD": false,
                    "LinkPartnerAdvertisedSpeedandDuplex1000THD": false,
                    "LinkPartnerAdvertisedSpeedandDuplex100TXFD": false,
                    "LinkPartnerAdvertisedSpeedandDuplex100TXHD": true,
                    "LinkPartnerAdvertisedSpeedandDuplex10TFD": false,
                    "LinkPartnerAdvertisedSpeedandDuplex10THD": true,
                    "OEMLinkSpeedValid": false,
                    "ParallelDetectionFlag": false,
                    "RXFlowControlFlag": false,
                    "SerDesLink": false,
                    "SpeedAndDuplex": "Auto-negotiate Not Complete",
                    "TXFlowControlFlag": false
                },
                "MACAddress": "a0:36:9f:70:0f:7f",
                "NcsiStatistics": {
                    "AENsSent": 0,
                    "NCSICommandChecksumErrors": 0,
                    "NCSICommandTypeErrors": 0,
                    "NCSICommandsReceived": 8,
                    "NCSIControlPacketsDropped": 0,
```

```
                    "NCSIReceivePackets": 8,
                    "NCSITransmitPackets": 8
                },
                "NcsicontrollerPacketStatistics": {
                    "1024To1522ByteFramesReceived": null,
                    "1024To1522ByteFramesTransmitted": null,
                    "128To255ByteFramesReceived": null,
                    "128To255ByteFramesTransmitted": null,
                    "1523To9022ByteFramesReceived": null,
                    "1523To9022ByteFramesTransmitted": null,
                    "256To511ByteFramesReceived": null,
                    "256To511ByteFramesTransmitted": null,
                    "512To1023ByteFramesReceived": null,
                    "512To1023ByteFramesTransmitted": null,
                    "64ByteFramesReceived": null,
                    "64ByteFramesTransmitted": null,
                    "65To127ByteFramesReceived": null,
                    "65To127ByteFramesTransmitted": null,
                    "AlignmentErrors": 0,
                    "ControlFramesReceived": null,
                    "CountersClearedFromLastRead": 517116,
                    "ErrorJabberPacketsReceived": null,
                    "ErrorRuntPacketsReceived": null,
                    "ExcessiveCollisionFrames": null,
                    "FCSReceiveErrors": 0,
                    "FalseCarrierDetections": null,
                    "JabberPacketsReceived": null,
                    "LateCollisionFrames": null,
                    "MultipleCollisionTransmitFrames": 0,
                    "PauseXOFFFramesReceived": 0,
                    "PauseXOFFFramesTransmitted": 0,
                    "PauseXONFramesReceived": 0,
                    "PauseXONFramesTransmitted": 0,
                    "RuntPacketsReceived": null,
                    "SingleCollisionTransmitFrames": 0,
                    "TotalBroadcastPacketsReceived": 0,
                    "TotalBroadcastPacketsTransmitted": 0,
                    "TotalBytesReceived": null,
                    "TotalBytesTransmitted": null,
                    "TotalMulticastPacketsReceived": 0,
                    "TotalMulticastPacketsTransmitted": 0,
                    "TotalUnicastPacketsReceived": 0,
                    "TotalUnicastPacketsTransmitted": 0,
                    "ValidBytesReceived": null
                },
                "PassThroughStatistics": {
                    "RXPacketChannelStateErrors": null,
                    "RXPacketOversizedErrors": null,
                    "RXPacketUndersizedErrors": null,
                    "RXPacketsReceivedonLANInterface": 0,
                    "TXPacketChannelStateErrors": null,
                    "TXPacketOversizedErrors": null,
                    "TXPacketUndersizedErrors": null,
                    "TXPacketsDropped": 0,
                    "TXPacketsReceivedonNCSIInterface": 0,
                    "TotalRXPacketsDropped": 0
                }
            }
        ]
    }
```

2.15 Common metrics value comparison table

	NetworkAdapterMetrics	NetworkDeviceFunctionMetrics	PortMetrics	InsydeNcsiPackage
Total Bytes Received	RXBytes (Sum of all channels)	RXBytes	RXBytes	NcsicontrollerPacketStatistics.TotalBytesTransmitted
Total Bytes Transmitted	TXBytes (Sum of all channels)	TXBytes	TXBytes	NcsicontrollerPacketStatistics.TotalBytesReceived
Total Unicast Packets Received	RXUnicastFrames (Sum of all channels)	RXUnicastFrames	Ethernet.RXUnicastFrames	NcsicontrollerPacketStatistics.TotalUnicastPacketsReceived
Total Multicast Packets Received	RXMulticastFrames (Sum of all channels)	RXMulticastFrames	Ethernet.RXMulticastFrames	NcsicontrollerPacketStatistics.TotalMulticastPacketsReceived
Total Unicast Packets Transmitted	TXUnicastFrames (Sum of all channels)	TXUnicastFrames	Ethernet.TXUnicastFrames	NcsicontrollerPacketStatistics.TotalUnicastPacketsTransmitted
Total Multicast Packets Transmitted	TXMulticastFrames (Sum of all channels)	TXMulticastFrames	Ethernet.TXMulticastFrames	NcsicontrollerPacketStatistics.TotalMulticastPacketsTransmitted
FCS Receive Errors			Ethernet.RXFCSErrors	NcsicontrollerPacketStatistics.FCSReceiveErrors
Pause XON Frames Received			Ethernet.RXPauseXONFrames	NcsicontrollerPacketStatistics.PauseXONFramesReceived
Pause XOFF Frames Received			Ethernet.RXPauseXOFFFrames	NcsicontrollerPacketStatistics.PauseXOFFFramesReceived
Control Frames Received			Ethernet.RXPFCFrames	NcsicontrollerPacketStatistics.ControlFramesReceived
Pause XON Frames Transmitted			Ethernet.TXPauseXONFrames	NcsicontrollerPacketStatistics.PauseXONFramesTransmitted
Pause XOFF Frames Transmitted			Ethernet.TXPauseXOFFFrames	NcsicontrollerPacketStatistics.PauseXOFFFramesTransmitted
Single Collision Transmit Frames			Ethernet.TXSingleCollisions	NcsicontrollerPacketStatistics.SingleCollisionTransmitFrames
Multiple Collision Transmit Frames			Ethernet.TXMultipleCollisions	NcsicontrollerPacketStatistics.MultipleCollisionTransmitFrames
Frames Received	NCSIRXFrames	RXFrames	Ethernet.RXFrames	NcsicontrollerPacketStatistics.64ByteFramesReceived

	(Sum of all channels: 64-Byte Frames Received 65-127 Byte Frames Received 128-255 Byte Frames Received 256-511 Byte Frames Received 512-1023 Byte Frames Received 1024-1522 Byte Frames Received 1523-9022 Byte Frames Received)	(Sum of: 64-Byte Frames Received 65-127 Byte Frames Received 128-255 Byte Frames Received 256-511 Byte Frames Received 512-1023 Byte Frames Received 1024-1522 Byte Frames Received 1523-9022 Byte Frames Received	(Sum of: 64-Byte Frames Received 65-127 Byte Frames Received 128-255 Byte Frames Received 256-511 Byte Frames Received 512-1023 Byte Frames Received 1024-1522 Byte Frames Received 1523-9022 Byte Frames Received	NcsicontrollerPacketStatistics.65To127ByteFramesReceived NcsicontrollerPacketStatistics.128To255ByteFramesReceived NcsicontrollerPacketStatistics.256To511ByteFramesReceived NcsicontrollerPacketStatistics.512To1023ByteFramesReceived NcsicontrollerPacketStatistics.1024To1522ByteFramesReceived NcsicontrollerPacketStatistics.1523To9022ByteFramesReceived
Frames Transmitted	NCSITXFrames (Sum of all channels: 64-Byte Frames Transmitted 65-127 Byte Frames Transmitted 128-255 Byte Frames Transmitted 256-511 Byte Frames Transmitted	TXFrames (Sum of: 64-Byte Frames Transmitted 65-127 Byte Frames Transmitted 128-255 Byte Frames Transmitted 256-511 Byte Frames Transmitted	Ethernet.TXFrames (Sum of: 64-Byte Frames Transmitted 65-127 Byte Frames Transmitted 128-255 Byte Frames Transmitted 256-511 Byte Frames Transmitted	NcsicontrollerPacketStatistics.64ByteFramesTransmitted NcsicontrollerPacketStatistics.65To127ByteFramesTransmitted NcsicontrollerPacketStatistics.128To255ByteFramesTransmitted NcsicontrollerPacketStatistics.256To511ByteFramesTransmitted NcsicontrollerPacketStatistics.512To1023ByteFramesTransmitted NcsicontrollerPacketStatistics.1024To1522ByteFramesTransmitted

	512-1023 Byte Frames Transmitted 1024-1522 Byte Frames Transmitted 1523-9022 Byte Frames Transmitted	512-1023 Byte Frames Transmitted 1024-1522 Byte Frames Transmitted 1523-9022 Byte Frames Transmitted	512-1023 Byte Frames Transmitted 1024-1522 Byte Frames Transmitted 1523-9022 Byte Frames Transmitted	NcsicontrollerPacketStatistics.1523To9022ByteFramesTransmitted

2.16 InsydePCIeFunctions

URI: /redfish/v1/Systems/system/Oem/Insyde/PCIeFunctions

This schema displays the PCIe information that comes from BIOS. A device with the property "SubClass" set to "Ethernet controller" is classified as a network device.


```
{
   "@odata.id": "/redfish/v1/Systems/system/Oem/Insyde/PCIeFunctions/",
   "@odata.type": "#InsydePCIeFunctions.v1_0_0.InsydePCIeFunctions",
   "Description": "This schema defines supported PCIeFunctions on this System.",
   "Id": "PCIeFunctions",
   "InsydePCIeInfo": [
     {
       "BDF": "84:00.0",
       "Class": "Network controller",
       "ClassCode": "0x020000",
       "Component": "SoC",
       "Device": "I210 Gigabit Network Connection",
       "DeviceId": "0x1533",
       "Interface": null,
       "RevisionId": "0x03",
       "SubClass": "Ethernet controller",
       "SubSystem": "Ethernet Server Adapter I210-T1",
       "SubsystemId": "0x0001",
       "SubsystemVendorId": "0x8086",
       "Vendor": "Intel Corporation",
       "VendorId": "0x8086"
     }...
   ],
   "Name": "PCIeFunctions"
 }
```